

QU Yichen

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PROFILE

Data Science undergraduate passionate about bridging AI, data science, and business analysis to solve real-world problems through intelligent and data-driven decision-making.

EDUCATION

Hong Kong Baptist University | B.Sc., Business Computing and Data Analytics Sep 2023 – Present
CGPA: 3.83/4.0 | COURSEWORK: Machine Learning; Data Mining; Causal Inference; Data Structures & Algorithms; Data Visualization; Linear Algebra; Economics
HONORS: APEC & HKSAR Government Scholarships; Outstanding Student Award; President’s Honour Roll

Nanyang Technological University | Exchange, Computer Science Aug 2025 – Dec 2025
COURSEWORK: Artificial Intelligence; Machine Learning; Visual & Predictive Analytics | SCHOLARSHIP: Albert and Mabel Chan Scholarship

EXPERIENCE

Business Operations & Management Intern, ByteDance May 2026 – Present

- Developed enterprise SQL pipelines and automated BI dashboards, migrating Excel-based workflows to support RMB 1B+ global IT budget allocation.
- Built cost allocation and forecasting models using time-series analysis and business rules to support AI infrastructure and workforce planning.
- Designed an LLM-based AI Agent integrating Rule Engine, LLM evaluation, and HITL for financial auditing, anomaly detection, and report automation.
- Standardized analytics methodologies and AI evaluation frameworks, improving operational efficiency by 5–8× and enabling scalable decision-making.

Project Assistant & AlphaAI Robotics Workshop Designer, HKBU Jun 2025 – Aug 2025

- Developed Python robotics demonstrations integrating computer vision and motion control.
- Designed AI learning modules covering neural networks and reinforcement learning.
- Optimized robot control scripts and communicated technical concepts to multidisciplinary audiences.

International Project Operator, Beijing Hong Kong Web3.0 Industry Center Jul 2024 – Aug 2024

- Supported cross-border startup operations, financial budgeting, and project coordination.
- Conducted data-driven marketing analysis and operational performance reviews.

PROJECTS

NanoGPT-Math: LLM Fine-Tuning with Direct Preference Optimization
Project Leader Singapore, Aug 2025 – Nov 2025

- Built a PyTorch pipeline for DPO fine-tuning with a 100k-pair preference dataset.
- Implemented custom loss balancing for stable optimization during LLM training.
- Evaluated mathematical reasoning performance through benchmark testing and error analysis.

Credit Risk Modeling with Machine Learning
Project Leader Singapore, Sep 2025 – Dec 2025

- Developed a Python ML pipeline for credit default prediction using transactional feature engineering.
- Benchmarked Logistic Regression, LightGBM, and XGBoost with K-fold cross-validation.
- Identified behavioral risk factors through model interpretation to support credit policy decisions.

SKILLS

AI: AI Agent, PyTorch, DPO, Neural Networks, RL **Data:** Python, SQL, Scikit-learn, XGBoost
Analytics: Modeling, Feature Engineering, Power BI **Tools:** Git/GitHub, Pandas, Matplotlib, LaTeX
Languages: English (IELTS 7.0), Mandarin (Native)